

## Ex:13 Network Address Translation (NAT)

### AIM:

To configure Network Address Translation (NAT) on a router to allow devices with private IP addresses to access external resources using a public IP.

### ALGORITHM:

#### 1. Set Up the Network:

- Drag and drop a **Router**, **Switch**, and **PCs** into the workspace.
- Connect the router to both the internal network (via a switch) and the external network (using a server to simulate the Internet).

#### 2. Configure Router Interfaces:

- Assign IP addresses to the router's internal and external interfaces:

- **Internal (Private IP):** 192.168.1.1/24
- **External (Public IP):** 209.165.200.1/24

Example:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown

Router(config)# interface gigabitEthernet0/1
Router(config-if)# ip address 209.165.200.1 255.255.255.0
Router(config-if)# no shutdown
```

#### 3. Define Access List for NAT:

- Create an access list to permit the private IP range to be translated by NAT.

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
```

#### 4. Enable NAT on the Router:

- Configure NAT on the router, mapping the private IP range to the public IP interface.

```
Router(config)# ip nat inside source list 1 interface
gigabitEthernet0/1 overload
```

#### 5. Configure Inside and Outside Interfaces:

Mark the router's internal interface as **inside** and the external interface as **outside**.

Example:

```
Router(config)# interface gigabitEthernet0/0
Router(config-if)# ip nat inside

Router(config)# interface gigabitEthernet0/1
```

```
Router(config-if)# ip nat outside
```

6. **Test the NAT Configuration:**

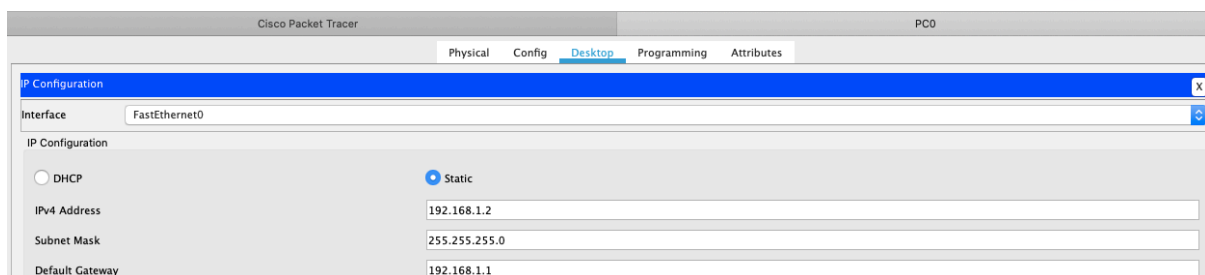
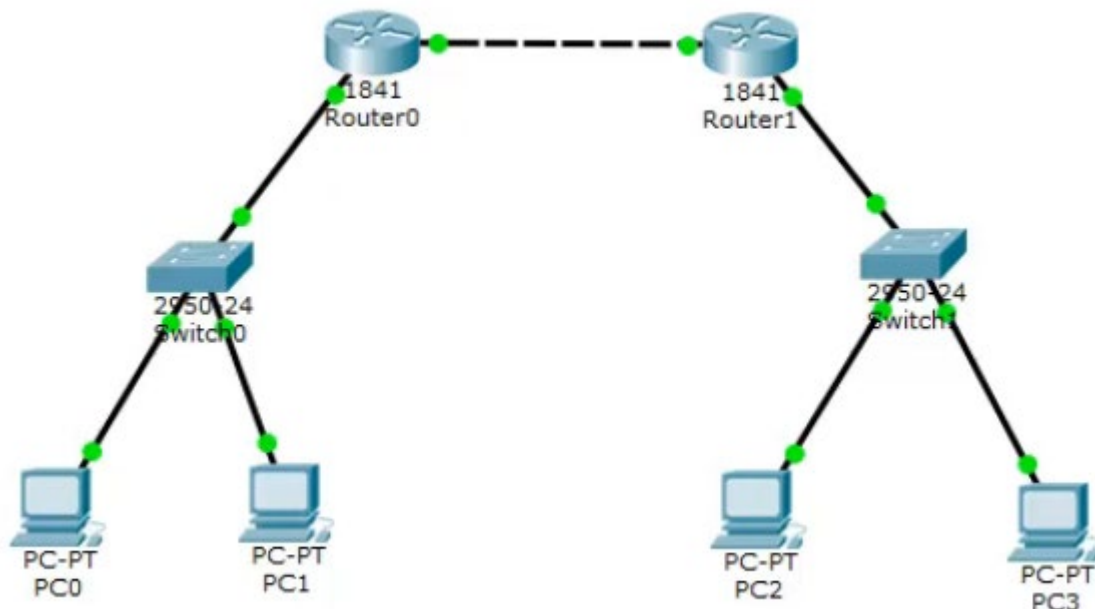
- From a PC in the internal network, attempt to access the server (Internet).
- Use **ping** or open a web page to test connectivity.

7. **Verify NAT Translations:**

- Check the NAT translations on the router:

```
Router# show ip nat translations
```

**OUTPUT:**



The screenshot displays three instances of the Cisco Packet Tracer interface, each showing the IP Configuration window for a different PC (PC1, PC2, and PC3). The configuration is as follows:

| Interface     | IP Configuration | IPv4 Address | Subnet Mask   | Default Gateway | DNS Server |
|---------------|------------------|--------------|---------------|-----------------|------------|
| FastEthernet0 | Static           | 192.168.1.3  | 255.255.255.0 | 192.168.2.1     | 0.0.0.0    |
| FastEthernet0 | Static           | 192.168.2.2  | 255.255.255.0 | 192.168.2.1     | 0.0.0.0    |
| FastEthernet0 | Static           | 192.168.2.3  | 255.255.255.0 | 192.168.2.1     | 0.0.0.0    |

Below the configuration windows, a Command Prompt window is open, showing the results of a ping test:

```
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254
Reply from 10.10.10.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

## RESULT:

- The NAT configuration successfully allows devices with private IP addresses in the internal network to communicate with the external network using the router's public IP.
- The **show ip nat translations** command shows the active NAT translations between private and public IP addresses.
- Successful **ping** responses or web access from internal PCs to external servers confirm that NAT is working as expected.